Methods results and limitations

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Methods results and limitations before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Finding primary research

Without opening the old lesson, explain one key idea from Finding primary research. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

A question that evidence can answer

Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 15

Practise with fading help

> Complete the supported Methods results and limitations trace and record the first step you can explain without help.

Use W2: Trace the worked case for **Method**s **result**s and **limitation**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 15

Individual reasoning check

> Check the proposed Methods results and limitations explanation against the supplied facts

> before discussing it.

Use the worked case for **Method**s **result**s and **limitation**s. Proposed explanation: “The worked **result** establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Statements:fixed cutoff5; 12/15 relevant returned items; judgments came from one reviewer. Classify.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Methods results and limitations facts and keep your reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Methods results and limitations facts and keep your reasoning.

Statements:used20 fictional queries; 16 passed; real student use was not studied. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Methods results and limitations facts and keep your reasoning.

A report omits this qualification: A **limitation** should constrain a claim rather than be hidden in unrelated prose. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

WORKBOOK · 8 OF 15

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Methods results and limitations facts and keep your reasoning.

Statements:fixed cutoff5; 12/15 relevant returned items; judgments came from one reviewer. Classify.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Methods results and limitations facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Methods results and limitations to Reproducible AI Research Study, identify a limit and choose the next useful test.

Add a labelled **method**s **result**s and **limitation**s **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **method**s **result**s and **limitation**s.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and **limitation**. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

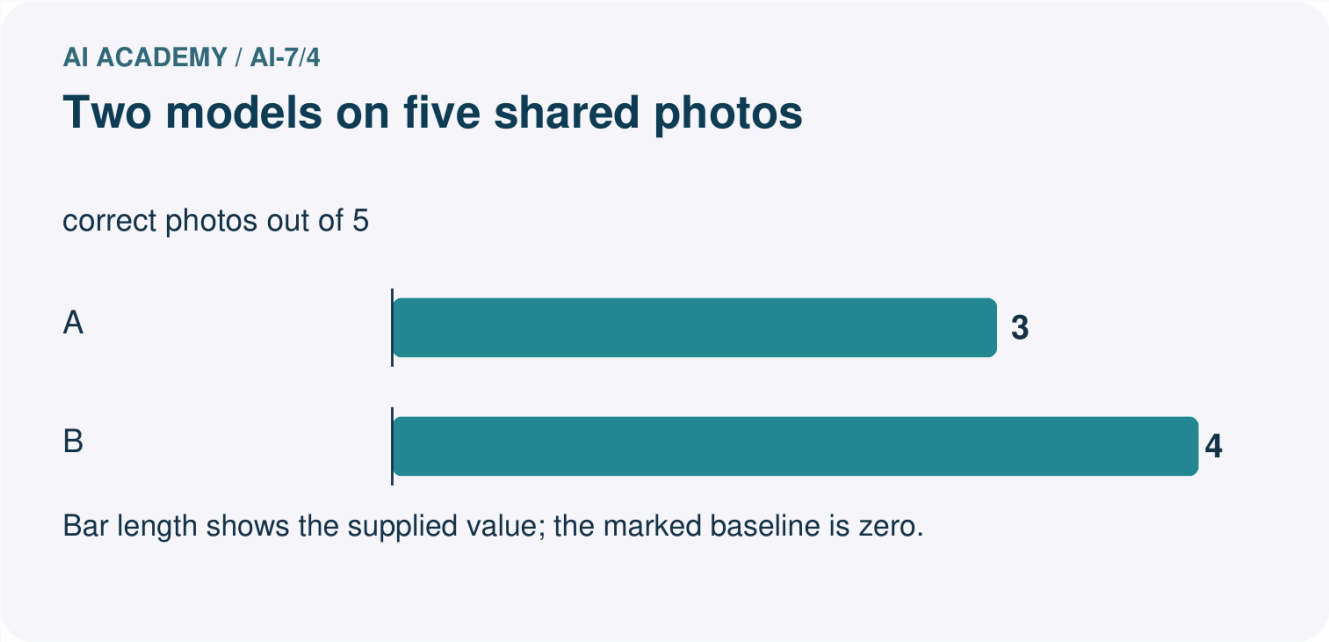
Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Pictorial case practice: Methods results and limitations

> Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera. Explain the labelled step before checking the changed condition.

Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera.

A is correct on 3/5; B on 4/5. No outdoor photos appear in the set.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/4

Methods results and limitations

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Identify the method

Same five photos, labels and scoring are used for both models.



2. State the result

A scores 60%; B scores 80% on this fixed set.



3. State the limit

The set supplies no outdoor-camera performance evidence.

CHANGE ONE THING

The conclusion says B is best for outdoor farming.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories. Why cannot the **result** establish that B is best for outdoor farming?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning.

Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Reproducible AI Research Study record.

Project: Reproducible AI Research Study

Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting **method**.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? The set supplies no outdoor-camera performance **evidence**.

1. State the **result**

2. State the limit

3. Identify the **method**

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: State the limit The set supplies no outdoor-camera performance **evidence**.

Methods tell how **evidence** was produced; **result**s state **observation**s; **limitation**s describe what those **observation**s cannot settle. Keeping them distinct prevents a **result** from becoming an oversized claim.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The conclusion says B is best for outdoor farming. Which conclusion is supported by the supplied facts?

1. Reject that extension without outdoor **evidence**.

2. This one example guarantees correct **result**s for every future **input**.

3. We can decide the **result** without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Reject that extension without

outdoor **evidence**. The reported conditions do not include the proposed use population.

Repair a missing building block

Review Finding primary research (ai-7/3). Goal: Explain the **mechanism** and terminology of finding primary research. Short practice: Separate what the blog claims from what the original report records.

Which **method**s and denominator must be inspected, and what status is honest if the original report is unavailable?. Return to this week and explain what you changed.

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

Explain each Methods results and limitations picture from its labels, then record a relevant

limit.

01. Methods results and limitations: the case

AI ACADEMY / AI-7/4 · 01 · LOOK AT THE FACTS

Methods results and limitations: the case

The situation

Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera.

Supplied evidence

A is correct on 3/5; B on 4/5. No outdoor photos appear in the set.

Explain methods results and limitations using the records and boundaries of this fictional investigation.

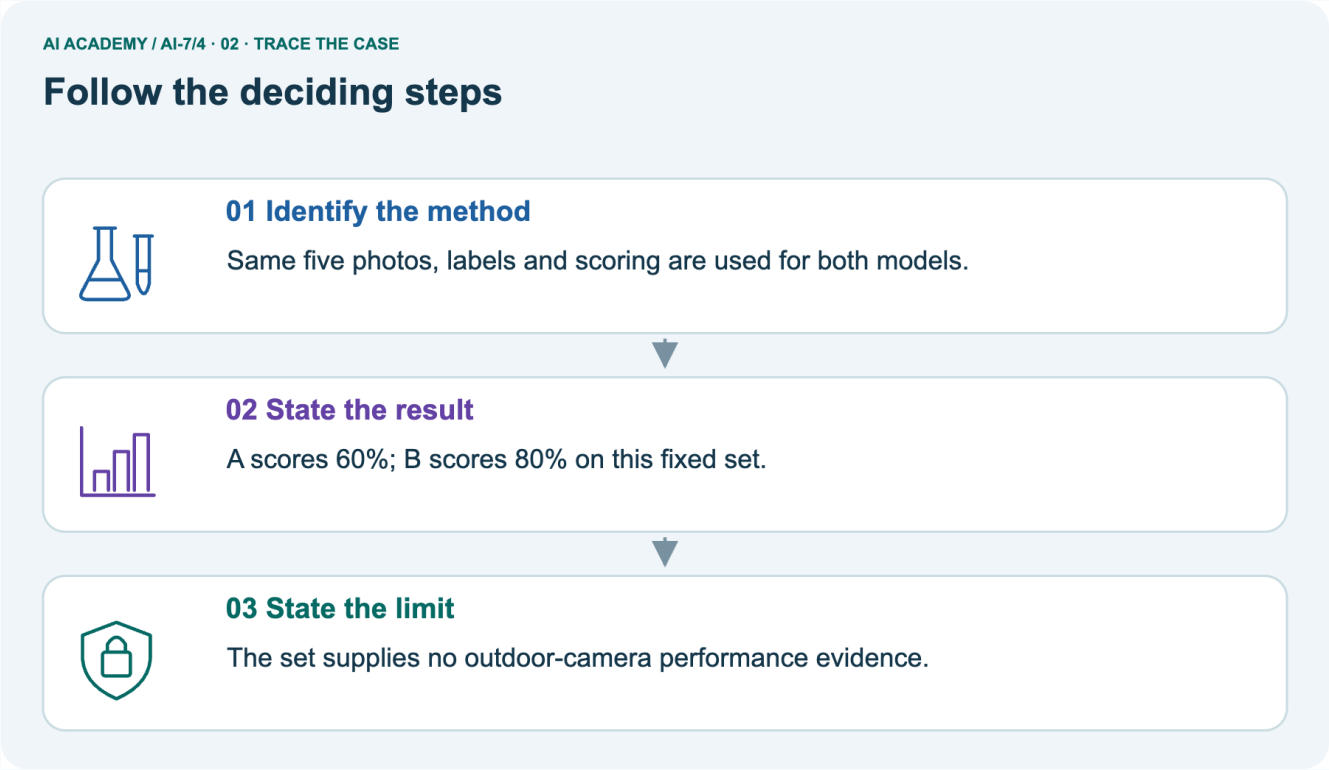
Follow the labelled picture with the matching explanation. Open the vector diagram

Separate procedure, finding and limit. A fictional report tests two frozen classifiers on five labelled leaf photos from one camera. Supplied facts: A is correct on 3/5; B on 4/5. No outdoor photos appear in the set.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



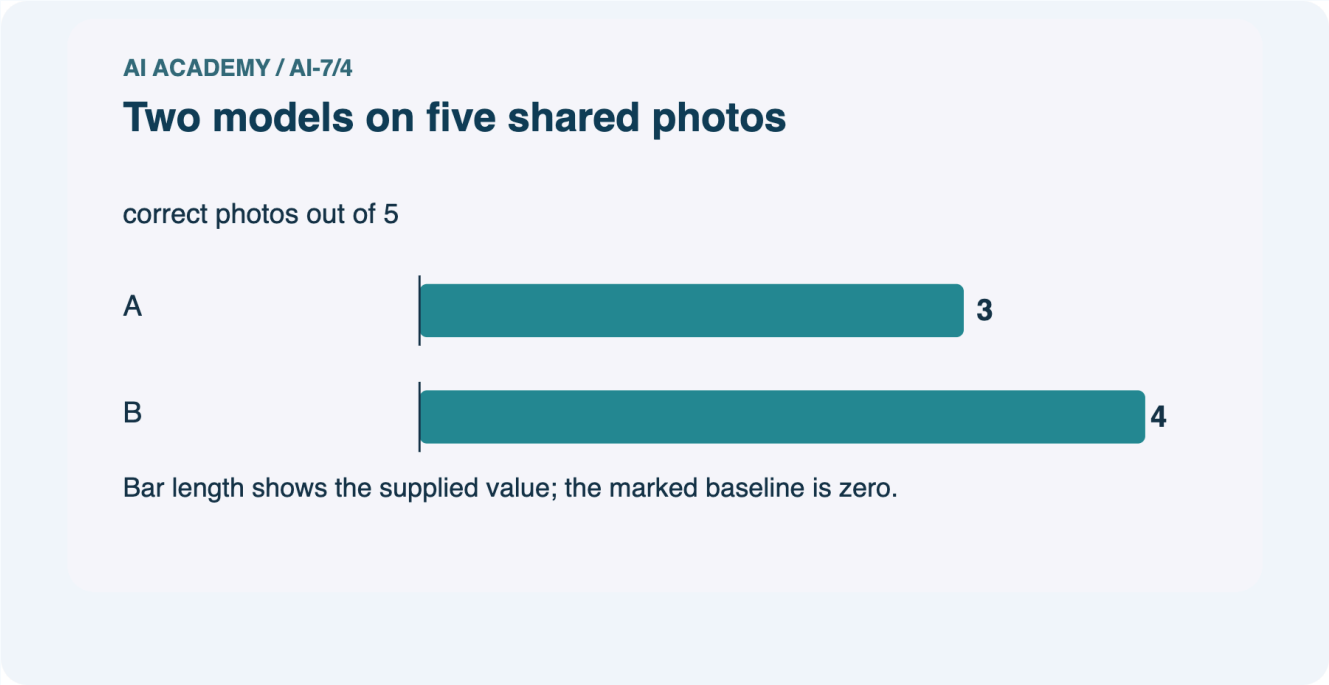
Follow the labelled picture with the matching explanation. Open the vector diagram

Identify the **method**: Same five photos, labels and scoring are used for both models. → State the **result**: A scores **60%**; B scores **80%** on this fixed set. → State the limit: The set supplies no outdoor-camera performance **evidence**.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Two models on five shared photos



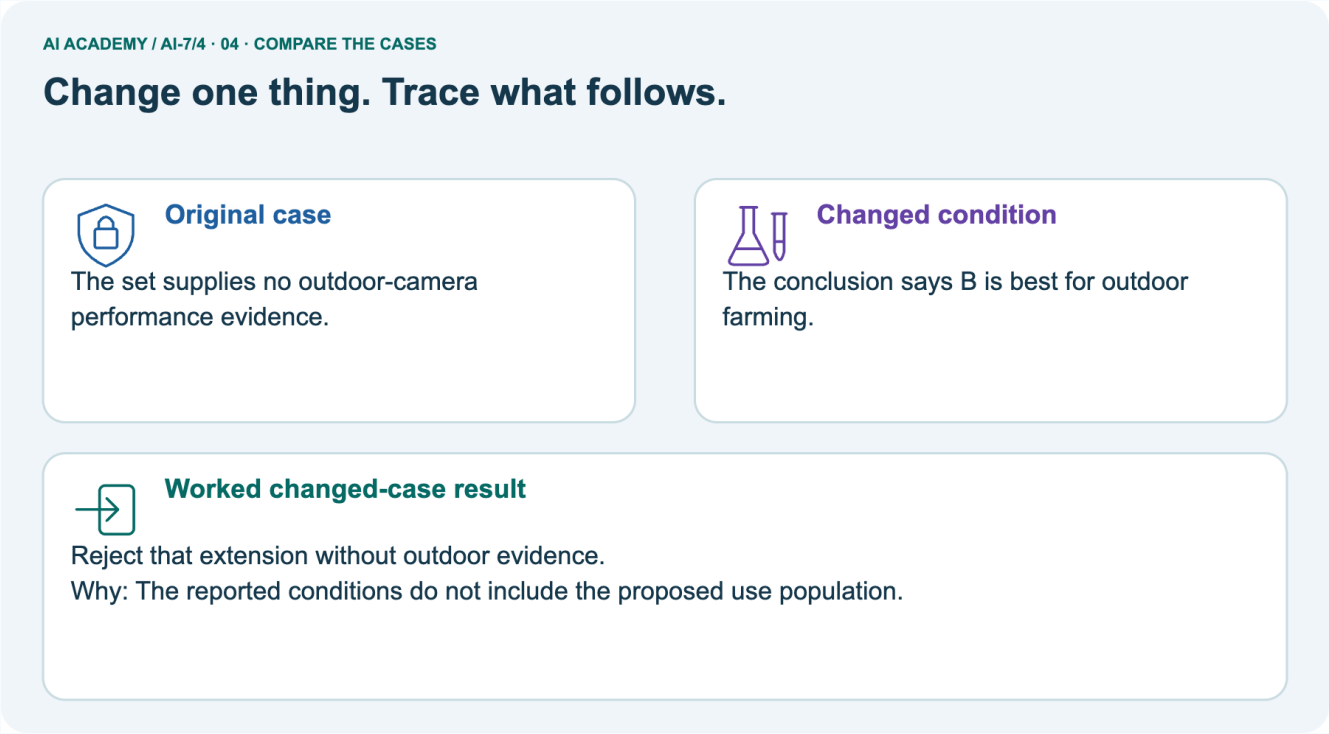
Follow the labelled picture with the matching explanation. Open the vector diagram

Two models on five shared photos. Units: correct photos out of 5. A: 3. B: 4.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

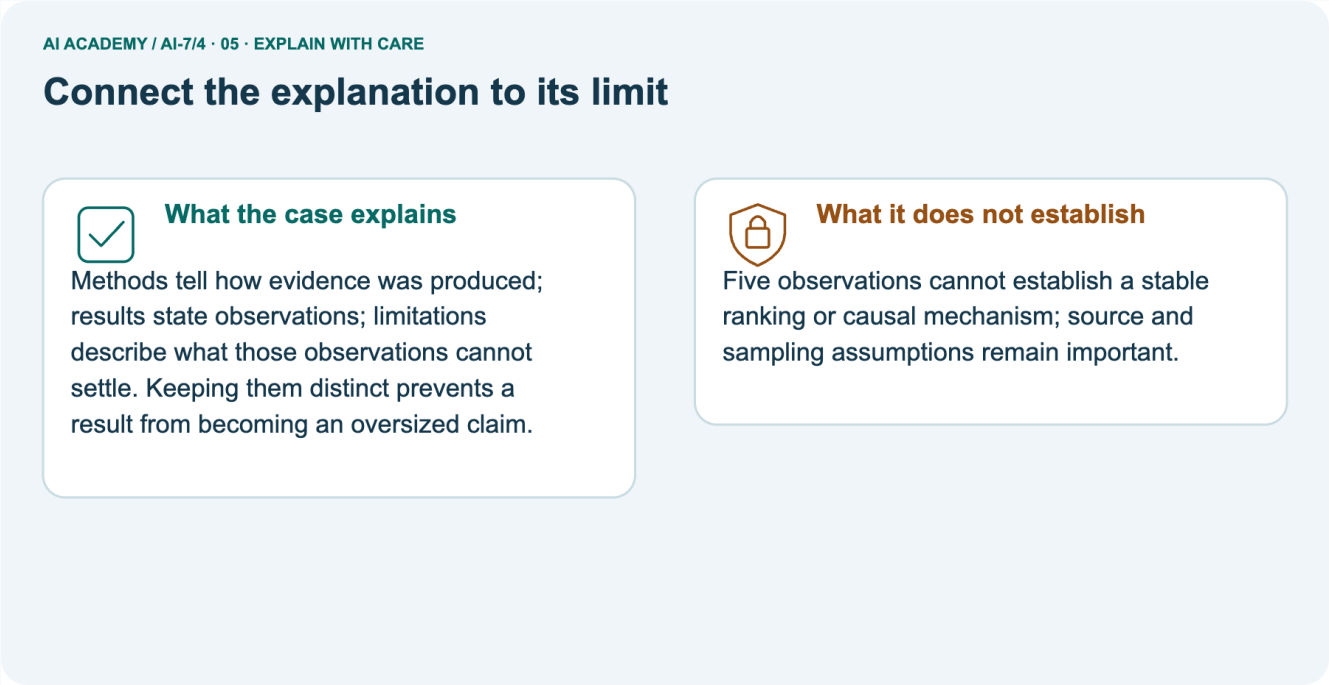
Original case: The set supplies no outdoor-camera performance **evidence**. Changed condition: The conclusion says B is best for outdoor farming. Expected **result**: Reject that extension without outdoor **evidence**.

Reason: The reported conditions do not include the proposed use population.

> Predict before checking the worked changed-case **result**. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

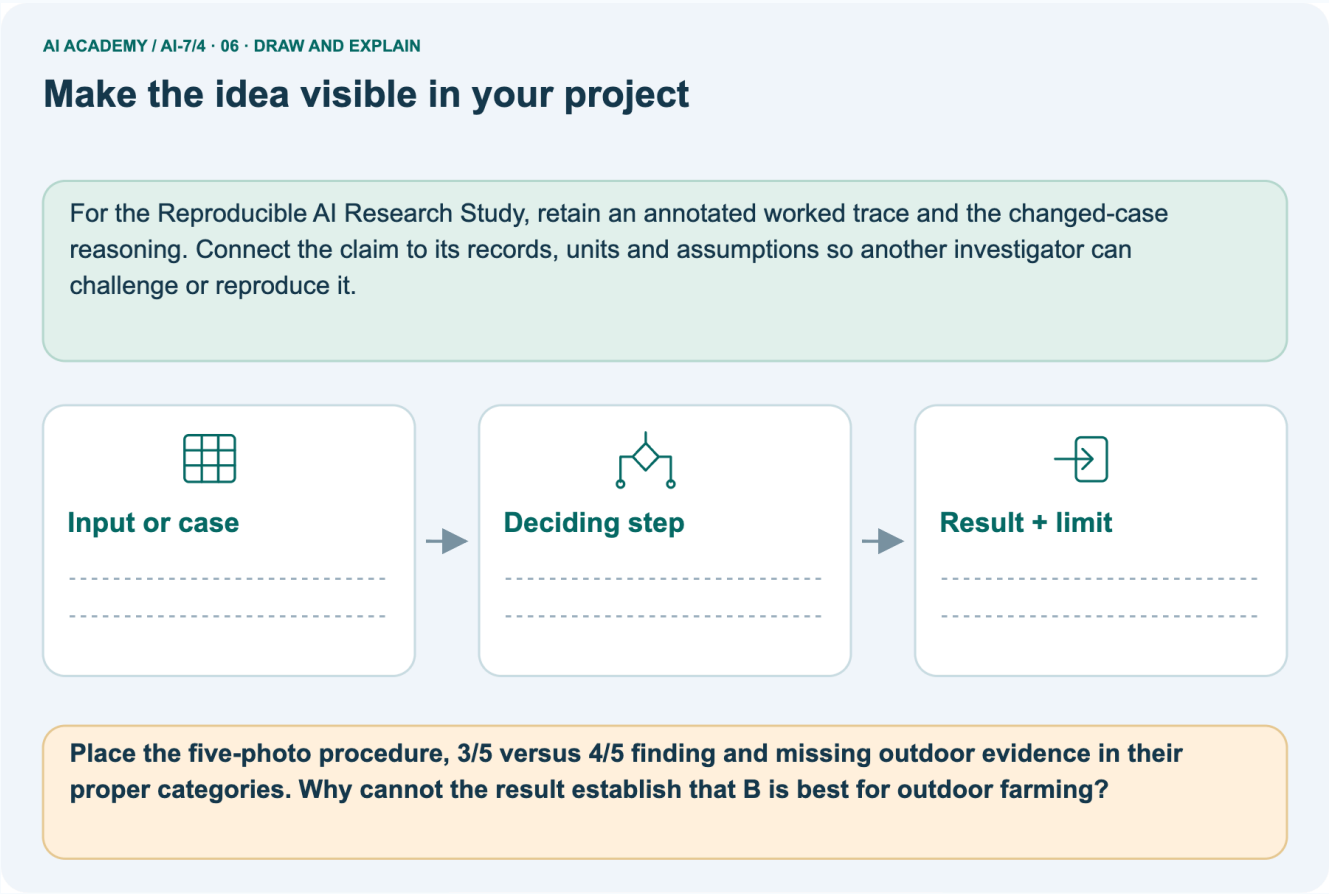
Methods tell how **evidence** was produced; **result**s state **observation**s; **limitation**s describe what those **observation**s cannot settle. Keeping them distinct prevents a **result** from becoming an oversized claim.

Limit: Five **observation**s cannot establish a stable ranking or causal **mechanism**; source and sampling assumptions remain important.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, **result** and limit. Supported question: Place the five-photo procedure, 3/5 versus 4/5 finding and missing outdoor **evidence** in their proper categories.

Why cannot the **result** establish that B is best for outdoor farming?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.